

tf.gather Stay organized with collections

Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/gather

Gather slices from params axis axis according to indices. (deprecated arguments)

Function Signatures

```
tf.gather( params, indices, validate_indices=None, axis=None,
batch_dims=0, name=None )
(validate_indices)
```

Code Examples

```
tf.gather(  
    params, indices, validate_indices=None, axis=None, batch_dims=0,  
    name=None  
)
```

```
tf.gather(  
    params, indices, validate_indices=None, axis=None, batch_dims=0,  
    name=None  
)
```

```
params = tf.constant(['p0', 'p1', 'p2', 'p3', 'p4', 'p5'])  
params[3].numpy()  
b'p3'  
tf.gather(params, 3).numpy()  
b'p3'
```

```
params = tf.constant(['p0', 'p1', 'p2', 'p3', 'p4', 'p5'])
```

```
params[3].numpy()
```

```
tf.gather(params, 3).numpy()
```

Documentation

Gather slices from `params` `axis` `axis` according to `indices`. (deprecated arguments)

Used in the notebooks

- Introduction to graphs and `tf.function`
 - Introduction to tensor slicing
 - Extension types
 - BERT Preprocessing with TF Text
 - Subword tokenizers
 - Simple audio recognition: Recognizing keywords
 - Image segmentation
 - Multilevel Modeling Primer in TensorFlow Probability
 - MoViNet for streaming action recognition
 - Client-efficient large-model federated learning via ``federated_select`` and sparse aggregation

Gather slices from `params` `axis` according to `indices`. `indices` must be an integer tensor of any dimension (often 1-D).

`Tensor.getitem` works for scalars, `tf.newaxis`, and python slices

`tf.gather` extends indexing to handle tensors of indices.

In the simplest case it's identical to scalar indexing:

The most common case is to pass a single axis tensor of indices (this can't be expressed as a python slice because the indices are not sequential):

The indices can have any shape. When the `params` has 1 axis, the output shape is equal to the input shape:

The params may also have any shape. `gather` can select slices across any axis depending on the `axis` argument (which defaults to 0). Below it is used to gather first rows, then columns from a matrix:

More generally: The output shape has the same shape as the input, with the indexed-axis replaced by the shape of the indices.

Here are some examples:

This is because each index takes a slice from `params`, and places it at the corresponding location in the output. For the above example

Batching:

The `batch_dims` argument lets you gather different items from each element of a batch.

Using `batch_dims=1` is equivalent to having an outer loop over the first axis of `params` and indices:

This is equivalent to:

Higher values of `batch_dims` are equivalent to multiple nested loops over the outer axes of `params` and indices. So the overall shape function is

This comes up naturally if you need to use the indices of an operation like `tf.argsort`, or `tf.math.top_k` where the last dimension of the indices indexes into the last dimension of input, at the corresponding location. In this case you can use `tf.gather(values, indices, batch_dims=-1)`.

See also:

- `tf.Tensor.getitem`: The direct tensor index operation (`t[]`), handles scalars and python-slices `tensor[..., 7, 1:-1]`

- `tf.scatter`: A collection of operations similar to `__setitem__`

`(t[i] = x)`

- `tf.gather_nd`: An operation similar to `tf.gather` but gathers across

multiple axis at once (it can gather elements of a matrix instead of rows or columns)

- `tf.boolean_mask`, `tf.where`: Binary indexing.
- `tf.slice` and `tf.strided_slice`: For lower level access to the

implementation of `__getitem__`'s python-slice handling (`t[1:-1:2]`)

Returns